Tom Otero

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SENIOR SOFTWARE ENGINEER

Engineer with experience in software systems design, embedded Linux, and DevOps automation for real-world platforms. My background ranges from running integration tests for new hardware and drivers in autonomous vehicles to orchestrating high-complexity build systems for large-scale AI frameworks. I thrive in roles that prioritize first-principles thinking, deterministic and reproducible systems, and domain-agnostic problem solving.

Experience

Google (via Yoh) — Senior Build & CI/CD Engineer (2025–Present) *Los Angeles, CA*

- Architected and implemented the primary recursive superbuild framework for LiteRT-LM using CMake, successfully decoupling the project from its Bazel-centric origins to enable portable, cross-platform distribution.
- Engineered a modular orchestration layer to manage a high-complexity dependency graph (including Abseil, Protobuf, Flatbuffers, and SentencePiece), ensuring hermetic build lifecycles independent of host environments.
- Developed a sophisticated re-integration strategy utilizing aggregate targets and root shims to resolve systemic version conflicts and symbol collisions across disparate Google-internal libraries.
- Authored custom patcher and shim modules to perform build-time surgery, ensuring deterministic linking across diverse hardware backends, including Android (NPU/GPU), Linux, macOS, and Windows.
- Established systemic architectural standards and reusable CMake modules to maintain the conceptual integrity of the LiteRT-LM runtime as it scales for new AI primitives and hardware acceleration backends.

Rockstar Games — Associate Build & Release Engineer (2022–2025) *Manhattan, NY*

- Modernized legacy frontend asset delivery by building custom MSBuild tasks to compile SASS, inject CDN-backed paths via dependency injection, and deploy bundles to Akamai using a modular Python + MSBuild toolchain. This enabled resilient, environment-specific builds across staging, development, and production.
- Migrated deployment automation from internal repo scripts to modular Python packages published via pip, improving reproducibility, maintainability, and version control across environments.
- Established a cross-platform SSH tunnel between isolated, security-hardened networks, enabling artifact delivery between TeamCity and Windows Server.
- Reduced build overhead and improved scalability by containerizing legacy services with Docker and Compose, laying groundwork for future orchestration.
- Authored detailed internal documentation covering pipelines, build workflows, and troubleshooting processes used for onboarding and production incident response.
- Coordinated production deployment troubleshooting with developers, leads, and NOC, ensuring minimal service disruption while maintaining communication discipline with stakeholders.

- Supported Agile teams operating in a 3-week sprint cycle, contributing to planning, retrospectives, and backlog grooming.

Cruise (via Novus Labs) — Embedded Linux Test Engineer (2020–2022) *San Francisco, CA (Remote)*

- Developed automation for hardware-in-the-loop (HIL) testbeds to validate embedded Linux drivers used in autonomous vehicle platforms.
- Integrated Jenkins and Buildkite CI/CD pipelines with HIL testbeds, enabling automated validation of drivers across distributed lab hardware.
- Maintained CI/CD pipelines using Jenkins, Buildkite, and JFrog Artifactory to coordinate reproducible embedded Linux driver builds, track artifacts, and deploy test runs across HIL infrastructure.
- Engineered hardware test validation into Git feature branch merge requests using custom webhooks and worker services, triggering real-time HIL jobs across embedded devices.
- Developed Python-based Pytest automation to validate driver features and edge cases, integrating APIs, writing custom wrappers, and stimulating systems via external lab hardware.
- Investigated and triaged C/C++ driver failures in collaboration with Cruise developers using serial logs, dmesg, Pytest traces, and synchronized debugging across distributed testbeds.

Novus Labs — Mechanical Engineer (2018–2022) *Hillsboro, OR*

- Collaborated with RF and electrical engineers to architect a distributed testbed for Amazon Alexa, using RF chambers, programmable attenuators, noise generators, and power cycling to simulate real-world interference and signal loss scenarios in compliance with RTCA DO-160G Section 20.
- Unified disparate device protocols into a test automation framework, coordinating Telnet-controlled power supplies, SSH-managed attenuators, USB-based CLI interfaces, and Ellisys Bluetooth sniffers.
- Coordinated timing in a distributed ATmega sensor and logging system using an NTP-aligned controller and custom synchronization protocol.
- Developed an automated A/V latency validation testbed for smart speakers, integrating a Raspberry Pi–based TTS module (Google API, ffmpeg, sox), light sensors, and servo-triggered actuators to replace manual input and validate command-response timing.

Education

Portland State University — B.S., Mechanical Engineering (2015–2019)

- Focused on machine design, control theory, and robotics.
- Completed projects in system modeling, feedback control, and inverse kinematics, including programming the path planning and inverse kinematics logic for a 6DOF robotic arm capable of writing letters in 3D space.
- Coursework included PID controller design, Kalman filters, and filter design in both digital and continuous domains using MATLAB and op-amp circuits.